

FLORIDA ENERGY EFFICIENCY CODE FOR BUILDING CONSTRUCTION

Florida Department of Business and Professional Regulation - Residential Performance Method

Project Name: Model Sappire		Builder Name: Arielle Construction	
Street: TBD SW 24th PI		Permit Office:	
City, State, Zip: Ocala, FL, 34481		Permit Number:	
Owner:		Jurisdiction:	
Design Location: FL, Ocala		County: Marion(Florida Climate Zone 2)	

1. New construction or existing	New (From Plans)	10. Wall Types(1548.0 sqft.)	Insulation	Area
2. Single family or multiple family	Detached	a. Concrete Block - Int Insul, Exterior	R=4.1	1416.00 ft ²
3. Number of units, if multiple family	1	b. Frame - Wood, Adjacent	R=13.0	132.00 ft ²
4. Number of Bedrooms	3	c. N/A		
5. Is this a worst case?	No	d. N/A		
6. Conditioned floor area above grade (ft ²)	1172	11. Ceiling Types(1172.0 sqft.)	Insulation	Area
Conditioned floor area below grade (ft ²)	0	12. Flat ceiling under att (Vented)	R=30.0	1172.00 ft ²
7. Windows(190.4 sqft.)	Description	13. N/A		
a. U-Factor:	Dbl, U=0.34	14. N/A		
SHGC:	SHGC=0.32	12. Roof(Comp. Shingles, Vented) Deck	R=0.0	1270 ft ²
b. U-Factor:	Dbl, U=0.63	13. Ducts, location & insulation level	R	ft ²
SHGC:	SHGC=0.32	a. Sup: Attic, Ret: Attic, AH: Great/Din		6 467
c. U-Factor:	N/A	b.		
SHGC:		c.		
Area Weighted Average Overhang Depth:	3.464 ft	14. Cooling Systems	kBtu/hr	Efficiency
Area Weighted Average SHGC:	0.320	a. Central Unit	28.4	SEER2:14.30
8. Skylights	Description	Area		
U-Factor:(AVG)	N/A	15. Heating Systems	kBtu/hr	Efficiency
SHGC(AVG):	N/A	a. Electric Heat Pump	28.4	HSPF2:7.50
9. Floor Types	Insulation	Area		
a. Slab-On-Grade Edge Insulation	R= 0.0	1172.00 ft ²		
b. N/A	R=	ft ²		
c. N/A	R=	ft ²		
		16. Hot Water Systems		
		a. Electric	Cap: 50 gallons	
			EF: 0.950	
		b. Conservation features		
			None	
		17. Credits	Pstat	

Glass/Floor Area: 0.163

Total Proposed Modified Loads: 37.11

Total Baseline Loads: 40.99

PASS

NOTE: Proposed residence must have annual total normalized Modified Loads that are less than or equal to 95 percent of the annual total loads of the standard reference design in order to comply.

I hereby certify that the plans and specifications covered by this calculation are in compliance with the Florida Energy Code.

PREPARED BY: Sten LullDATE: 6/26/2024

I hereby certify that this building, as designed, is in compliance with the Florida Energy Code.

OWNER/AGENT: Donato MosqueraDATE: 6/26/24

Review of the plans and specifications covered by this calculation indicates compliance with the Florida Energy Code. Before construction is completed this building will be inspected for compliance with Section 553.908 Florida Statutes.



BUILDING OFFICIAL: _____

DATE: _____

- Compliance requires certification by the air handler unit manufacturer that the air handler enclosure qualifies as certified factory-sealed in accordance with R403.3.2.1.
- Default duct leakage does not require a Duct Leakage Test Report.
- Compliance requires an Air Barrier and Insulation Inspection Checklist in accordance with R402.4.1.1 and this project requires a PERFORMANCE envelope leakage test report with envelope leakage no greater than 7.00 ACH50 (R402.4.1.2).

Reviewed for Code Compliance

2024064281

6/26/2024 8:31:33 AM

EnergyGauge® USA 8.0.000 - Florida 2023 FBC 8th Edition (2023) Compliant Software

OFFICE COPY
TO REMAIN IN OFFICE
UNTIL JOB IS FINALED

Plans have not been reviewed for
Mechanical, Electrical, or Plumbing

INPUT SUMMARY CHECKLIST REPORT

PROJECT										
Title: Model Sapphire				Address type: Street Address						
Building Type: User	Bedrooms: 3			Lot #: ---						
Owner:	Conditioned Area: 1172			Block/SubDivision: ---						
Builder Home ID:	Total Stories: 1			PlatBook: ---						
Builder Name: Arielle Construction	Worst Case: No			Street: TBD SW 24th PI						
Permit Office:	Rotate Angle: 0			County: Marion						
Jurisdiction:	Cross Ventilation: No			City, State, Zip: Ocala,						
Family Type: Detached	Whole House Fan: No			FL, 34481						
New/Existing: New (From Plans)	Terrain: Suburban									
Year Construct:	Shielding: Suburban									
Comment:										
CLIMATE										
✓ Design Location	Tmy Site	Design Temp 97.5%	Design Temp 2.5%	Int Design Temp Winter	Int Design Temp Summer	Heating Degree Days	Design Moisture	Daily temp Range		
___ FL, Ocala	FL_OCALA_MUNI_(AWOS)	28	91	70	75	1144.5	51	Medium		
BLOCKS										
✓ Number	Name	Area	Volume							
___ 1	Entire House	1172	10530 cu ft							
SPACES										
✓ Number	Name	Area	Volume	Kitchen	Occupants	Bedrooms	Finished	Cooled	Heated	
___ 1	Great/Dine/Kit	430	3870	Yes	0		Yes	Yes	Yes	
___ 2	Laund	37	333	No	0		Yes	Yes	Yes	
___ 3	M Bedroom	191	1719	No	2	1	Yes	Yes	Yes	
___ 4	M Bath	77	693	No	0		Yes	Yes	Yes	
___ 5	M Wic	61	549	No	0		Yes	Yes	Yes	
___ 6	Bath	45	405	No	0		Yes	Yes	Yes	
___ 7	Bedroom 2	149	1341	No	1	1	Yes	Yes	Yes	
___ 8	Bedroom 3	153	1377	No	1	1	Yes	Yes	Yes	
___ 9	WC	27	243	No	0		Yes	Yes	Yes	
FLOORS (Total Exposed Area = 1170 sq.ft.)										
✓ #	Floor Type	Space	Exposed Perim(ft)	Area	R-Value Perim. Joist	U-Factor	Slab Insul. Vert/Horiz	Tile	Wood	Carpet
___ 1	Slab-On-Grade Edge Ins	Great/Dine/Kit	32	429.6 sqft	---	0.304	2 (ft)/0 (ft)	0.00	0.00	1.00
___ 2	Slab-On-Grade Edge Ins	Laund	15.3	37.3 sqft	---	0.304	2 (ft)/0 (ft)	0.00	0.00	1.00
___ 3	Slab-On-Grade Edge Ins	M Bedroom	34.3	190.7 sqft	---	0.304	2 (ft)/0 (ft)	0.00	0.00	1.00
___ 4	Slab-On-Grade Edge Ins	M Bath	16	77.3 sqft	---	0.304	2 (ft)/0 (ft)	0.00	0.00	1.00
___ 5	Slab-On-Grade Edge Ins	M Wic	7.7	61.3 sqft	---	0.304	2 (ft)/0 (ft)	0.00	0.00	1.00
___ 6	Slab-On-Grade Edge Ins	Bath	5.7	45.3 sqft	---	0.304	2 (ft)/0 (ft)	0.00	0.00	1.00
___ 7	Slab-On-Grade Edge Ins	Bedroom 2	25	148.6 sqft	---	0.304	2 (ft)/0 (ft)	0.00	0.00	1.00
___ 8	Slab-On-Grade Edge Ins	Bedroom 3	32.7	153.2 sqft	---	0.304	2 (ft)/0 (ft)	0.00	0.00	1.00
___ 9	Slab-On-Grade Edge Ins	WC	3.3	26.7 sqft	---	0.304	2 (ft)/0 (ft)	0.00	0.00	1.00

INPUT SUMMARY CHECKLIST REPORT

ROOF												
✓ #	Type	Materials	Roof Area	Gable Area	Roof Color	Rad Barr	Solar Absor.	SA Tested	Emitt	Emitt Tested	Deck Insul.	Pitch (deg)
1	Hip	Composition shingles	1270 ft²	0 ft²	Medium	N	0.9	No	0.9	No	0	22.62

ATTIC						
✓ #	Type	Ventilation	Vent Ratio (1 in)	Area	RBS	IRCC
1	Full attic	Vented	300	1172 ft²	N	N

CEILING (Total Exposed Area = 1170 sq.ft.)								
✓ #	Ceiling Type	Space	R-Value	Ins. Type	Area	U-Factor	Framing Frac.	Truss Type
1	Flat ceiling under attic(Vented)	Great/Dine/Kit	30.0	Blown	430.0ft²	0.053	0.10	Wood
2	Flat ceiling under attic(Vented)	Laund	30.0	Blown	37.0ft²	0.053	0.10	Wood
3	Flat ceiling under attic(Vented)	M Bedroom	30.0	Blown	191.0ft²	0.053	0.10	Wood
4	Flat ceiling under attic(Vented)	M Bath	30.0	Blown	77.0ft²	0.053	0.10	Wood
5	Flat ceiling under attic(Vented)	M Wic	30.0	Blown	61.0ft²	0.053	0.10	Wood
6	Flat ceiling under attic(Vented)	Bath	30.0	Blown	45.0ft²	0.053	0.10	Wood
7	Flat ceiling under attic(Vented)	Bedroom 2	30.0	Blown	149.0ft²	0.053	0.10	Wood
8	Flat ceiling under attic(Vented)	Bedroom 3	30.0	Blown	153.0ft²	0.053	0.10	Wood
9	Flat ceiling under attic(Vented)	WC	30.0	Blown	27.0ft²	0.053	0.10	Wood

WALLS (Total Exposed Area = 1548 sq.ft.)													
✓ #	Omt	Adjacent To	Wall Type	Space	Cavity R-Value	Width Ft In	Height Ft In	Area sq.ft.	U-Factor	Sheath R-Value	Frm. Frac.	Solar Absor.	Below Grade
1	N	Exterior	Conc. Blk - Int Ins	Great/Dine/Kit	4.1	19.0 0	9.0 0	171.0	0.149	0	0	0.45	0 %
2	S	Exterior	Conc. Blk - Int Ins	Great/Dine/Kit	4.1	13.0 0	9.0 0	117.0	0.149	0	0	0.45	0 %
3	S	Exterior	Conc. Blk - Int Ins	Laund	4.1	7.0 4	9.0 0	66.0	0.149	0	0	0.45	0 %
4	W	Exterior	Conc. Blk - Int Ins	Laund	4.1	4.0 0	9.0 0	36.0	0.149	0	0	0.45	0 %
5	E	Garage	Frame - Wood	Laund	13.0	4.0 0	9.0 0	36.0	0.095	0	0.25	0.45	0 %
6	N	Exterior	Conc. Blk - Int Ins	M Bedroom	4.1	12.0 0	9.0 0	108.0	0.149	0	0	0.45	0 %
7	E	Exterior	Conc. Blk - Int Ins	M Bedroom	4.1	13.0 4	9.0 0	120.0	0.149	0	0	0.45	0 %
8	W	Exterior	Conc. Blk - Int Ins	M Bedroom	4.1	9.0 0	9.0 0	81.0	0.149	0	0	0.45	0 %
9	E	Exterior	Conc. Blk - Int Ins	M Bath	4.1	5.0 4	9.0 0	48.0	0.149	0	0	0.45	0 %
10	S	Garage	Frame - Wood	M Bath	13.0	10.0 8	9.0 0	96.0	0.095	0	0.25	0.45	0 %
11	E	Exterior	Conc. Blk - Int Ins	M Wic	4.1	7.0 8	9.0 0	69.0	0.149	0	0	0.45	0 %
12	W	Exterior	Conc. Blk - Int Ins	Bath	4.1	5.0 8	9.0 0	51.0	0.149	0	0	0.45	0 %
13	N	Exterior	Conc. Blk - Int Ins	Bedroom 2	4.1	14.0 0	9.0 0	126.0	0.149	0	0	0.45	0 %
14	W	Exterior	Conc. Blk - Int Ins	Bedroom 2	4.1	11.0 0	9.0 0	99.0	0.149	0	0	0.45	0 %
15	E	Exterior	Conc. Blk - Int Ins	Bedroom 3	4.1	7.0 4	9.0 0	66.0	0.149	0	0	0.45	0 %
16	S	Exterior	Conc. Blk - Int Ins	Bedroom 3	4.1	14.0 0	9.0 0	126.0	0.149	0	0	0.45	0 %
17	W	Exterior	Conc. Blk - Int Ins	Bedroom 3	4.1	11.0 4	9.0 0	102.0	0.149	0	0	0.45	0 %
18	E	Exterior	Conc. Blk - Int Ins	WC	4.1	3.0 4	9.0 0	30.0	0.149	0	0	0.45	0 %

DOORS (Total Exposed Area = 20 sq.ft.)										
✓ #	Omt	Adjacent To	Door Type	Space	Storms	U-Value	Width Ft In	Height Ft In	Area	
1	S	Exterior	Insulated	Great/Dine/Kit	None	0.60	3.00 0	6.00 8	20.0ft²	

INPUT SUMMARY CHECKLIST REPORT

WINDOWS															(Total Exposed Area = 190 sq.ft.)			
✓ #	Omt	Wall ID	Frame	Panes	NFRC U-Factor	SHGC	Imp	Storm	Total Area (ft²)	Same Units	Width (ft)	Height (ft)	—Overhang— Depth (ft)	Sep. (ft)	Interior Shade	Screen		
1	N	1	Vinyl	Low-E Double	Y	0.34	0.32	N	N	13.3	1	2.67	5.00	2.0	2.2	Drapes/blinds	None	
2	N	1	Metal	Low-E Double	Y	0.63	0.32	N	N	40.0	1	6.00	6.67	8.8	0.8	Drapes/blinds	None	
3	S	2	Vinyl	Low-E Double	Y	0.34	0.32	N	N	25.8	1	5.00	5.17	2.0	2.2	Drapes/blinds	None	
4	S	3	Vinyl	Low-E Double	Y	0.34	0.32	N	N	5.3	1	2.00	2.67	3.7	2.7	Drapes/blinds	None	
5	N	3	Vinyl	Low-E Double	Y	0.34	0.32	N	N	17.8	1	2.67	6.67	2.0	2.2	Drapes/blinds	None	
6	N	6	Vinyl	Low-E Double	Y	0.34	0.32	N	N	25.8	1	5.00	5.17	2.0	2.2	Drapes/blinds	None	
7	E	9	Vinyl	Low-E Double	Y	0.34	0.32	N	N	5.3	1	2.00	2.67	2.0	2.2	Drapes/blinds	None	
8	W	12	Vinyl	Low-E Double	Y	0.34	0.32	N	N	5.3	1	2.00	2.67	2.0	2.2	Drapes/blinds	None	
9	N	13	Vinyl	Low-E Double	Y	0.34	0.32	N	N	25.8	1	5.00	5.17	2.0	2.2	Drapes/blinds	None	
10S		16	Vinyl	Low-E Double	Y	0.34	0.32	N	N	25.8	1	5.00	5.17	2.0	2.2	Drapes/blinds	None	

INFILTRATION										
✓ #	Scope	Method	SLA	CFM50	ELA	EqLA	ACH	ACH50	Space(s)	Infiltration Test Volume
1	Wholehouse	Proposed ACH(50)	0.00040	1229	67.40	126.53	0.1211	7.0	All	10530 cu ft

GARAGE					
✓ #	Floor Area	Roof Area	Exposed Wall Perimeter	Avg. Wall Height	Exposed Wall Insulation
1	231 ft²	231 ft²	49 ft	9 ft	0

MASS					
✓ #	Mass Type	Area	Thickness	Furniture Fraction	Space
1	Default(8 lbs/sq.ft.)	0 ft²	0 ft	0.30	Great/Dine/Kit
2	Default(8 lbs/sq.ft.)	0 ft²	0 ft	0.30	Laund
3	Default(8 lbs/sq.ft.)	0 ft²	0 ft	0.30	M Bedroom
4	Default(8 lbs/sq.ft.)	0 ft²	0 ft	0.30	M Bath
5	Default(8 lbs/sq.ft.)	0 ft²	0 ft	0.30	M Wic
6	Default(8 lbs/sq.ft.)	0 ft²	0 ft	0.30	Bath
7	Default(8 lbs/sq.ft.)	0 ft²	0 ft	0.30	Bedroom 2
8	Default(8 lbs/sq.ft.)	0 ft²	0 ft	0.30	Bedroom 3
9	Default(8 lbs/sq.ft.)	0 ft²	0 ft	0.30	WC

HEATING SYSTEM										
✓ #	System Type	Subtype/Speed	AHRI #	Efficiency	Capacity kBTu/hr	—Geothermal HeatPump—			Ducts	Block
						Entry	Power	Volt	Current	
1	Electric Heat Pump	Split/Single		HSPF2: 7.50	28.4		0.00	0.00	0.00	sys#1 1

COOLING SYSTEM									
✓ #	System Type	Subtype/Speed	AHRI #	Efficiency	Capacity kBTu/hr	Air Flow cfm	SHR	Duct	Block
1	Central Unit	Split/Single		SEER2:14.3	28.4	0	0.83	sys#1	1

INPUT SUMMARY CHECKLIST REPORT

HOT WATER SYSTEM													
✓ #	System Type	Subtype	Location	EF(UEF)	Cap	Use	SetPnt	Fixture Flow	Pipe Ins.	Pipe length			
___ 1	Electric	None	Garage	0.95 (0.93)	50.00 gal	61 gal	120 deg	Standard	None	78			
	Recirculation System	Recirc Control Type	Loop length	Branch length	Pump power	DWHR	Facilities Connected	Equal Flow	DWHR Eff	Other Credits			
___ 1	No		NA	NA	NA	No	NA	NA	NA	None			
DUCTS													
✓ Duct #	Supply Location	R-Value	Area	Return Location	R-Value	Area	Leakage Type	Air Handler	CFM 25 TOT	CFM 25 OUT	QN OUT RLF	HVAC # Heat Cool	
___ 1	Attic	6.0	467 ft²	Attic	6.0	467 ft²	Default Leakage	Great/Dine	(Default)	(Default)		1 1	
TEMPERATURES													
Programable Thermostat: Y						Ceiling Fans: N							
Cooling	☐ Jan	☐ Feb	☐ Mar	☐ Apr	☐ May	☒ Jun	☒ Jul	☒ Aug	☒ Sep	☐ Oct	☐ Nov	☐ Dec	
Heating	☒ Jan	☒ Feb	☒ Mar	☐ Apr	☐ May	☐ Jun	☐ Jul	☐ Aug	☐ Sep	☐ Oct	☒ Nov	☒ Dec	
Venting	☐ Jan	☐ Feb	☒ Mar	☒ Apr	☐ May	☐ Jun	☐ Jul	☐ Aug	☐ Sep	☒ Oct	☒ Nov	☐ Dec	
✓ Thermostat Schedule: HERS 2006 Reference	Schedule Type	1	2	3	4	5	6	7	8	9	10	11	12
___ Cooling (WD)	AM PM	78 80	78 80	78 78	78 78	78 78	78 78	78 78	78 78	80 78	80 78	80 78	80 78
___ Cooling (WEH)	AM PM	78 78	78 78	78 78	78 78	78 78	78 78	78 78	78 78	78 78	78 78	78 78	78 78
___ Heating (WD)	AM PM	66 68	66 68	66 68	66 68	66 68	68 68	68 68	68 68	68 68	68 68	68 66	68 66
___ Heating (WEH)	AM PM	66 68	66 68	66 68	66 68	66 68	68 68	68 68	68 68	68 68	68 68	68 66	68 66

ENERGY PERFORMANCE LEVEL (EPL) DISPLAY CARD

ESTIMATED ENERGY PERFORMANCE INDEX* = 91

The lower the EnergyPerformance Index, the more efficient the home.

TBD SW 24th Pl,Ocala,FL,34481

1. New construction or existing	New (From Plans)	10. Wall Types(1548.0 sqft.)	Insulation	Area
2. Single family or multiple family	Detached	a. Concrete Block - Int Insul, Exterior	R=4.1	1416.00 ft ²
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4. Number of Bedrooms	3	c. N/A		
5. Is this a worst case?	No	d. N/A		
6. Conditioned floor area above grade (ft ²)	1172	11. Ceiling Types(1172.0 sqft.)	Insulation	Area
Conditioned floor area below grade (ft ²)	0	12. Flat ceiling under att (Vented)	R=30.0	1172.00 ft ²
7. Windows**	Description	13. N/A		
a. U-Factor:	Dbl, U=0.34	14. N/A		
SHGC:	SHGC=0.32	12. Roof(Comp. Shingles, Vented)	Deck R=0.0	1270 ft ²
b. U-Factor:	Dbl, U=0.63	13. Ducts, location & insulation level	R	ft ²
SHGC:	SHGC=0.32	a. Sup: Attic, Ret: Attic, AH: Great/Din	6	467
c. U-Factor:	N/A	b.		
SHGC:		c.		
Area Weighted Average Overhang Depth:	3.464 ft	14. Cooling Systems	kBtu/hr	Efficiency
Area Weighted Average SHGC:	0.320	a. Central Unit	28.4	SEER2:14.30
8. Skylights	Description	Area		
U-Factor:(AVG)	N/A	N/A ft ²		
SHGC(AVG):	N/A			
9. Floor Types	Insulation	Area		
a. Slab-On-Grade Edge Insulation	R= 0.0	1172.00 ft ²		
b. N/A	R=	ft ²		
c. N/A	R=	ft ²		
		15. Heating Systems	kBtu/hr	Efficiency
		a. Electric Heat Pump	28.4	HSPF2:7.50
		16. Hot Water Systems		
		a. Electric	Cap: 50 gallons	
			EF: 0.950	
		b. Conservation features		
			None	
		17. Credits	Pstat	

I certify that this home has complied with the Florida Energy Efficiency Code for Building Construction through the above energy saving features which will be installed (or exceeded) in this home before final inspection. Otherwise, a new EPL Display Card will be completed based on installed Code compliant features.

Builder Signature: Donato Mosquera Date: 6/26/24

Address of New Home: TBD SW 24th Pl City/FL Zip: Ocala,FL,34481



*Note: This is not a Building Energy Rating. If your Index is below 70, your home may qualify for energy efficient mortgage (EEM) incentives if you obtain a Florida Energy Rating. For information about the Florida Building Code, Energy Conservation, contact the Florida Building Commission's support staff.

**Label required by Section R303.1.3 of the Florida Building Code, Energy Conservation, if not DEFAULT.

Load Short Form Entire House

Job: 1172 Sq Ft
Date: Feb 29, 2024
By: Steven Ferrell

Project Information

For: Model Sappire, Arielle Development Corp
Ocala, FL

Design Information

	Htg	Clg	Infiltration	
Outside db (°F)	34	92	Method	Simplified
Inside db (°F)	70	75	Construction quality	Semi-loose
Design TD (°F)	36	17	Fireplaces	0
Daily range	-	M		
Inside humidity (%)	50	50		
Moisture difference (gr/lb)	32	42		

HEATING EQUIPMENT

Make Carrier
Trade 15 SEER2 HP
Model GH5SAN43000AA0
AHRI ref 210998680
Efficiency 7.5 HSPF2
Heating input
Heating output 28600 Btuh @ 47°F
Temperature rise 30 °F
Actual air flow 875 cfm
Air flow factor 0.034 cfm/Btuh
Static pressure 0.50 in H2O
Space thermostat
Capacity balance point = 32 °F

Backup:
Input = 8 kW, Output = 27702 Btuh, 100 AFUE

COOLING EQUIPMENT

Make Carrier
Trade 15 SEER2 HP
Cond GH5SAN43000AA0
Coil FJ4DNXB30L
AHRI ref 210998680
Efficiency 12.0 EER2, 14.3 SEER2
Sensible cooling 23572 Btuh
Latent cooling 4828 Btuh
Total cooling 28400 Btuh
Actual air flow 875 cfm
Air flow factor 0.037 cfm/Btuh
Static pressure 0.50 in H2O
Load sensible heat ratio 0.88

ROOM NAME	Area (ft²)	Htg load (Btuh)	Clg load (Btuh)	Htg AVF (cfm)	Clg AVF (cfm)
Great/Dine/Kit	430	6332	13956	216	514
Laund	37	2072	1364	71	50
M Bedroom	191	4888	2490	167	92
M Bath	77	1735	794	59	29
M Wc	61	1061	413	36	15
Bath	45	836	637	29	23
Bedroom 2	149	3640	1774	124	65
Bedroom 3	153	4625	2135	158	79
WC	27	461	180	16	7

Calculations approved by ACCA to meet all requirements of Manual J 8th Ed.



wrightsoft®
A McCook® / Danaher Performance Company

Right-Suite® Universal 2024 24.0.01 RSU22433
...velopment OcalaModel SappireModel Sappire.rup Calc = MJ8

2024-Jun-25 17:53:17

Page 1

Entire House	d	1172	25650	23743	875	875
Other equip loads			0	0		
Equip. @ 0.96 RSM				22912		
Latent cooling				3264		
TOTALS		1172	25650	26176	875	875

Calculations approved by ACCA to meet all requirements of Manual J 8th Ed.



Building Analysis Entire House

Job: 1172 Sq Ft
Date: Feb 29, 2024
By: Steven Ferrell

Project Information

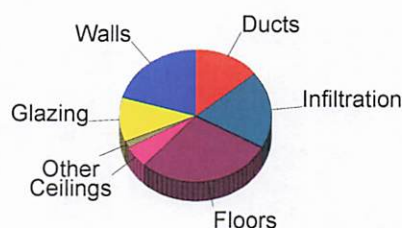
For: Model Sappire, Arielle Development Corp
Ocala, FL

Design Conditions

Location:		Indoor:		Heating	Cooling
Ocala Intl J Taylor, FL, US		Indoor temperature (°F)		70	75
Elevation: 89 ft		Design TD (°F)		36	17
Latitude: 29°N		Relative humidity (%)		50	50
		Moisture difference (gr/lb)		32.1	42.4
Outdoor:		Heating	Cooling		
Dry bulb (°F)		34	92		
Daily range (°F)		-	18 (M)		
Wet bulb (°F)		-	75		
Wind speed (mph)		15.0	7.5		
		Infiltration:			
		Method		Simplified	
		Construction quality		Semi-loose	
		Fireplaces		0	

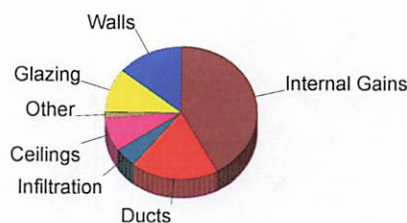
Heating

Component	Btuh/ft²	Btuh	% of load
Walls	3.9	5262	20.5
Glazing	14.6	2772	10.8
Doors	21.8	436	1.7
Ceilings	1.2	1359	5.3
Floors	6.3	7367	28.7
Infiltration	3.5	4890	19.1
Ducts		3564	13.9
Piping		0	0
Humidification		0	0
Ventilation		0	0
Adjustments		0	0
Total		25650	100.0



Cooling

Component	Btuh/ft²	Btuh	% of load
Walls	2.5	3362	14.2
Glazing	13.3	2535	10.7
Doors	17.3	347	1.5
Ceilings	1.7	1962	8.3
Floors	0	0	0
Infiltration	0.8	1143	4.8
Ducts		4374	18.4
Ventilation		0	0
Internal gains		10020	42.2
Blower		0	0
Adjustments		0	0
Total		23743	100.0



Latent Cooling Load = 3264 Btuh

Overall U-value = 0.122 Btuh/ft²·°F, Window / Floor Area = 16.3 %

Data entries checked.



Component Constructions Entire House

Job: 1172 Sq Ft
Date: Feb 29, 2024
By: Steven Ferrell

Project Information

For: Model Sapphire, Arielle Development Corp
Ocala, FL

Design Conditions

Location:		Indoor:		Heating	Cooling
Ocala Intl J Taylor, FL, US		Indoor temperature (°F)		70	75
Elevation: 89 ft		Design TD (°F)		36	17
Latitude: 29°N		Relative humidity (%)		50	50
		Moisture difference (gr/lb)		32.1	42.4
Outdoor:	Heating	Cooling			
Dry bulb (°F)	34	92			
Daily range (°F)	-	18 (M)			
Wet bulb (°F)	-	75			
Wind speed (mph)	15.0	7.5			
		Infiltration:			
		Method	Simplified		
		Construction quality	Semi-loose		
		Fireplaces	0		

Construction descriptions

	Area ft²	U-value Btuh/ft²·°F	Insul R ft²·°F/Btuh	Htg HTM Btuh/ft²	Loss Btuh	Clg HTM Btuh/ft²	Gain Btuh
Walls							
Blk wall, stucco ext, r-4 ext bd ins, 8" thk, 1/2" gypsum board int fnsh	300	0.110	4.1	3.99	1198	2.63	789
	328	0.110	4.1	3.99	1308	2.63	861
	232	0.110	4.1	3.99	926	2.63	610
	364	0.110	4.1	3.99	1452	2.63	956
	1223	0.110	4.1	3.99	4885	2.63	3216
Partitions							
12C-0sw: Fm wall, r-13 cav ins, 1/2" gypsum board int fnsh, 2"x4" wood frm, 16" o.c. stud	114	0.091	13.0	3.30	377	1.27	146
Windows							
4A5-2ov: 2 glazing, dr low-e outr, air gas, vnl frm mat, dr innr, 1/4" gap, 1/4" thk; NFRC rated (SHGC=0.32); 50% blinds 45°, medium; foreground = green grass (0.23); 2 ft overhang (5 ft window ht, 2.17 ft sep.); 6.67 ft head ht	13	0.340	0	12.3	165	10.7	142
4A5-2ov: 2 glazing, dr low-e outr, air gas, vnl frm mat, dr innr, 1/4" gap, 1/4" thk; NFRC rated (SHGC=0.32); 50% blinds 45°, medium; foreground = green grass (0.23); 2 ft overhang (5.17 ft window ht, 2.17 ft sep.); 6.67 ft head ht	52	0.340	0	12.3	638	10.7	552
	52	0.340	0	12.3	638	12.7	655
	103	0.340	0	12.3	1275	11.7	1207
4A5-2ovd: 2 glazing, dr low-e outr, air gas, vnl frm mat, dr innr, 1/4" gap, 1/4" thk; NFRC rated (SHGC=0.32); 50% blinds 45°, medium; foreground = new concrete (0.32); 8.75 ft overhang (6.67 ft window ht, 0.83 ft sep.); 6.67 ft head ht	40	0.630	0	22.9	915	15.5	618
4A5-2ov: 2 glazing, dr low-e outr, air gas, vnl frm mat, dr innr, 1/4" gap, 1/4" thk; NFRC rated (SHGC=0.32); 50% blinds 45°, medium; foreground = green grass (0.23); 2 ft overhang (2.67 ft window ht, 2.17 ft sep.); 6.67 ft head ht	5	0.340	0	12.3	66	34.4	183
	5	0.340	0	12.3	66	34.4	183
	11	0.340	0	12.3	132	34.4	367
4A5-2ov: 2 glazing, dr low-e outr, air gas, vnl frm mat, dr innr, 1/4" gap, 1/4" thk; NFRC rated (SHGC=0.32); 50% blinds 45°, medium; foreground = green grass (0.23); 3.67 ft overhang (2.67 ft window ht, 2.67 ft sep.); 6.67 ft head ht	5	0.340	0	12.3	66	12.7	68

10D-v: 2 glazing, dr low-e outr, air gas, vnl frm mat, dr innr, 1/4" gap, 1/4" thk; NFRC rated (SHGC=0.32); 50% blinds 45°, medium; foreground = new concrete (0.32); 2 ft overhang (6.67 ft window ht, 2.17 ft sep.); 6.67 ft head ht

18 0.340 0 12.3 220 7.47 133

Doors

11J0: Door, mtl fbrgl type

20 0.600 6.3 21.8 436 17.3 347

Ceilings

16B-30ad: Attic ceiling, asphalt shingles roof mat, r-30 ceil ins, 1/2" gypsum board int fnsh

1172 0.032 30.0 1.16 1359 1.68 1962

Floors

22A-tpm: Bg floor, heavy dry or light damp soil, on grade depth, carp 80% flr fnsh

172 1.180 0 42.8 7367 0 0

Duct System Summary

Entire House

Job: 1172 Sq Ft
Date: Feb 29, 2024
By: Steven Ferrell

Project Information

For: Model Sappire, Arielle Development Corp
Ocala, FL

	Heating	Cooling
External static pressure	0.50 in H2O	0.50 in H2O
Pressure losses	0.31 in H2O	0.31 in H2O
Available static pressure	0.19 in H2O	0.19 in H2O
Supply / return available pressure	0.136 / 0.054 in H2O	0.136 / 0.054 in H2O
Lowest friction rate	0.085 in/100ft	0.085 in/100ft
Actual air flow	875 cfm	875 cfm
Total effective length (TEL)	223 ft	

Supply Branch Detail Table

Name	Design (Btuh)	Htg (cfm)	Clg (cfm)	Design FR	Diam (in)	H x W (in)	Duct Matl	Actual Ln (ft)	Ftg.Eqv Ln (ft)	Trunk
Bath	h 836	29	23	0.086	4.0	0x0	VIFx	23.6	135.0	st1
Bedroom 2	h 3640	124	65	0.088	6.0	0x0	VIFx	25.4	130.0	st1
Bedroom 3	h 4625	158	79	0.085	7.0	0x0	VIFx	25.0	135.0	st1
Great/Dine/Kit-B	c 4652	72	171	0.097	7.0	0x0	VIFx	15.4	125.0	st2
Great/Dine/Kit-C	c 4652	72	171	0.092	7.0	0x0	VIFx	17.9	130.0	st1
Great/Dine/Kit-D	c 4652	72	171	0.090	7.0	0x0	VIFx	21.1	130.0	st1
Laund	h 2072	71	50	0.095	5.0	0x0	VIFx	17.9	125.0	st2
MBath	h 1735	59	29	0.098	4.0	0x0	VIFx	14.4	125.0	st2
M Bedroom-A	h 4888	167	92	0.094	7.0	0x0	VIFx	20.3	125.0	st2
M Wc	h 1061	36	15	0.097	4.0	0x0	VIFx	16.0	125.0	st2
WC	h 461	16	7	0.098	4.0	0x0	VIFx	13.8	125.0	st2

Supply Trunk Detail Table

Name	Trunk Type	Htg (cfm)	Clg (cfm)	Design FR	Veloc (fpm)	Diam (in)	H x W (in)	Duct Material	Trunk
st1	Peak AVF	454	510	0.085	650	12.0	0 x 0	VinIFlx	
st2	Peak AVF	421	365	0.094	535	12.0	0 x 0	VinIFlx	

Return Branch Detail Table

Name	Grille Size (in)	Htg (cfm)	Clg (cfm)	TEL (ft)	Design FR	Veloc (fpm)	Diam (in)	H x W (in)	Stud/Joist Opening (in)	Duct Matl	Trunk
rb3	0x 0	158	79	62.9	0.086	590	7.0	0x 0		VIFx	rt1
rb2	0x 0	124	65	63.3	0.085	632	6.0	0x 0		VIFx	rt1
rb4	10x 13	278	143	52.4	0.103	629	9.0	0x 0		VIFx	
rb1	16x 17	315	588	59.3	0.091	749	12.0	0x 0		VIFx	rt1

Return Trunk Detail Table

Name	Trunk Type	Htg (cfm)	Clg (cfm)	Design FR	Veloc (fpm)	Diam (in)	H x W (in)	Duct Material	Trunk
rt1	PeakAVF	597	732	0.085	685	14.0	0 x 0	VinIFlx	

Manual S Compliance Report

Entire House

Job: 1172 Sq Ft
Date: Feb 29, 2024
By: Steven Ferrell

Project Information

For: Model Sappire, Arielle Development Corp
Ocala, FL

Cooling Equipment

Design Conditions

Outdoor design DB:	91.5°F	Sensible gain:	23743	Btuh	Entering coil DB:	76.3°F
Outdoor design WB:	75.4°F	Latent gain:	3264	Btuh	Entering coil WB:	63.1°F
Indoor design DB:	75.0°F	Total gain:	27007	Btuh		
Indoor RH:	50%	Estimated airflow:	875	cfm		

Manufacturer's Performance Data at Actual Design Conditions

Equipment type:	Split ASHP			
Manufacturer:	Carrier	Model:	GH5SAN43000AA0+FJ4DNXB30L	
Actual airflow:	875	cfm		
Sensible capacity:	22817	Btuh	96% of load	
Latent capacity:	4289	Btuh	131% of load	
Total capacity:	27107	Btuh	100% of load	SHR: 84%

Heating Equipment

Design Conditions

Outdoor design DB:	33.7°F	Heat loss:	25650	Btuh	Entering coil DB:	69.4°F
Indoor design DB:	70.0°F					

Manufacturer's Performance Data at Actual Design Conditions

Equipment type:	Split ASHP			
Manufacturer:	Carrier	Model:	GH5SAN43000AA0+FJ4DNXB30L	
Actual airflow:	875	cfm		
Output capacity:	23755	Btuh	93% of load	Capacity balance: 32 °F
Supplemental heat required:	1895	Btuh		Economic balance: -99 °F

Backup equipment type:	Elec strip			
Manufacturer:		Model:		
Actual airflow:	875	cfm		
Output capacity:	8.1	KW	108% of load	Temp. rise: 50 °F

Meets all requirements of ACCA Manual S.



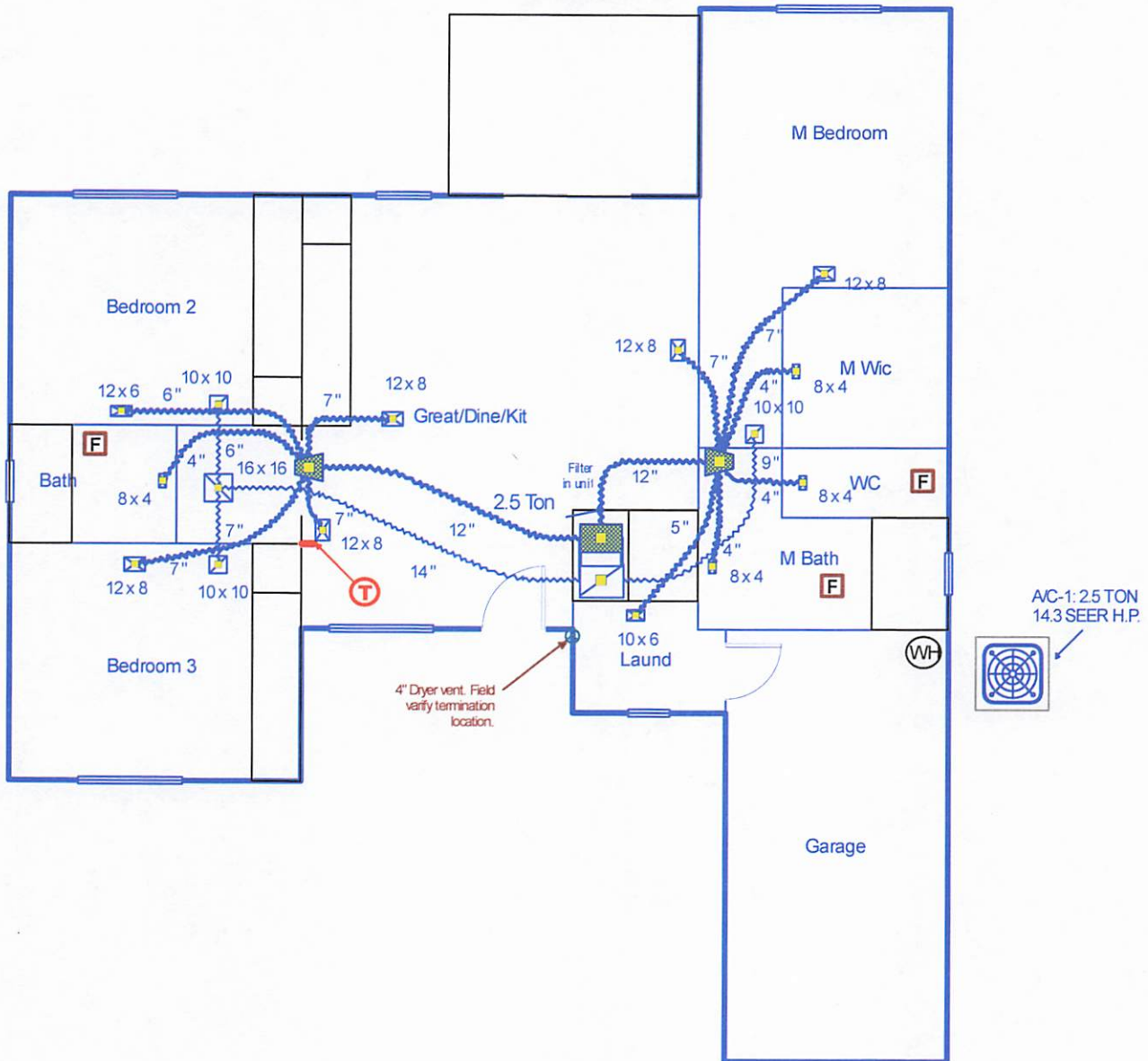
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Page 1

FIRST FLOOR



Job #: 1172 Sq Ft
 Performed by Steven Ferrell for:
 Model Sappire
 Ocala, FL

Scale: 1 : 92
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